

RESEARCH PRPOSAL



Propose Title of the Project:

Building a Web Application to Simplify Midwives' Daily Tasks in Maternal
and Newborn Care

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1.Introduction

Maternal and newborn health forms one of the most critical concerns in global health, especially for those countries with limited accesses to medical resources. Midwives represent the frontline workers in health service provision attending mothers and newborns, especially in the under-socially-economically strata of our communities. They are responsible for the observation of pregnancies, conducting prenatal and postnatal care, providing family planning services, attending deliveries, and instructing families on health practices. Despite the worth of the role they play, midwives often go through a number of challenges in the process of effectively discharging their duties. Major issues include high volumes of paperwork, difficulty in managing records of patients, and limited access to updated medical information. All these factors combine to increase the workload and stress of the midwives.

Besides, midwives are often found working in remotely located places with a scarcity of infrastructure in accessing real-time data, collaboration with other professionals, and keeping track of patients. Besides limiting midwives' productivity, these challenges reduce the quality of care provided and put the health and life of mothers and their newborns at risk. Over recent years, digital technology has emerged as a transforming solution in many areas of health.

For instance, web-based applications can facilitate, among others, administrative process streamlining, enhance communication, and improve the channels through which critical health information is accessed. Through web technologies, midwives can manage their day more wisely, access the latest information in real-time, and better collaborate with medical teams. If well-designed with midwives' purposes in mind, a web application can provide less manual effort and minimize mistakes while offering easy access to critical patient information, important for timely quality care. The next proposal undertakes simplification by designing and developing a web-based application to assist midwives in daily cares for mothers and newborns. The developed proposed solution should have all the features outlined below that would enable midwives to manage, track, remind, and schedule visits. It should allow midwives to securely manage patient records, access, and update them; track pregnancy and postpartum progress by using data entry forms with checklists; set reminders and schedule tasks to ensure timely antenatal and postnatal visits.

- Access to educational resources and updated guidelines on maternal and newborn health
- Real-time communication with doctors, nurses, and other healthcare providers for consultation or emergencies

This research will be guided by the principles of Human-Computer Interaction, which will make this application user-friendly and intuitive, tuned to the peculiar needs of midwives.

This will involve continuous solicitation in the design process from direct contributions of midwives, after which the tool will be developed for use by midwives to improve productivity by reducing administrative burdens. This will lead to an improved quality of maternal and newborn care, hence contributing to improved health outcomes and, subsequently, improved healthcare delivery in resource-constrained settings. The study, therefore, is aimed at equipping midwives with a digital tool that will ease the working process for better decision-making and coordination in maternal-newborn care. When developing this web application, usability and functionality will be in focus, hence allowing midwives to spend more time with patients and less time on administrative work.

2. Literature Revives

- The L10K 2020 project in Ethiopia successfully piloted and scaled an mHealth solution to enhance maternal and child health services. They created smartphone-based applications for Health Extension Workers and tablet-based apps for their supervisors and midwives, integrating modules for client registration, appointment management, follow-up notifications, job aids, referral, and tracking systems. Evaluations showed the apps were user-friendly and improved real-time data exchange, defaulter tracing, referrals, feedback, and adherence to care protocols. Effective stakeholder engagement was crucial for successful implementation, ownership, and skill transfer. The model for scaling requires careful analysis of resources, infrastructure, and potential challenges like mobile data capacity, power, and connectivity to ensure sustained impact.
Nigussie, Z. Y., Zemicheal, N. F., Tiruneh, G. T., Bayou, Y. T., Teklu, G. A., Kibret, E. S., ... & Betemariam, W. A. (2021). Using mHealth to improve timeliness and quality of maternal and newborn health in the primary health care system in Ethiopia. *Global Health: Science and Practice*, 9(3), 668-681.

- The midwife mobile phone project in Aceh Besar, Indonesia, showed that using mobile phones effectively enhanced healthcare delivery among 223 midwives across 15 health centers. Midwives used project phones to report health statistics, consult with coordinators, exchange health advice with peers, and communicate with doctors and patients, which facilitated quicker emergency responses and improved information flow to the patient community. Guided by the ICT for Healthcare Development and Technology-Community-Management models, the project revealed that mobile technology supported efficient healthcare communication but also faced challenges. Key barriers identified included infrastructure, economic limitations, technological constraints, and socio-cultural factors, underscoring the need for a balanced approach that combines top-down support with participatory, community-driven engagement.

Chib, A. (2010). The Aceh Besar midwives with mobile phones project: Design and evaluation perspectives using the information and communication technologies for healthcare development model. *Journal of Computer-Mediated Communication*, 15(3), 500-525.

- The DigiMAPS (Digitalization of Maternal and Post-natal Care System) project introduced an electronic medical records system (KIA Online) and a Tele-Obgyn tele-consultation platform aimed at improving maternal and child health services. This cloud-based platform supports both primary healthcare providers and inter-institutional access, including government and private facilities. Using an open-source design, DigiMAPS provides a user-friendly interface compatible with mobile devices, enhancing data management and visualization. Developed through lean innovation management, DigiMAPS incorporated rapid prototyping and user feedback from healthcare providers and authorities, adopting a participatory approach. The platform streamlines patient data management, reduces errors associated with paper-based records, and supports integration with broader healthcare systems. This centralized system simplifies administrative tasks and supports more accurate, real-time clinical data collection and reporting across various levels of the healthcare network.
- Faried, A., Sutiono, A. B., Djuwantono, T., Arifin, M. Z., Wirakusumah, F. F., Yuniarto, S. A., ... & Jayadi, Y. T. (2015, November). Mother and children health reporting system: Innovative information system application in the rural West

Bandung Area, Indonesia, by using multimodal communications systems. In 2015 4th International Conference on Instrumentation, Communications, Information Technology, and Biomedical Engineering (ICICI-BME) (pp. 202-207). IEEE.

- The field of midwifery has expanded to address the comprehensive health needs of women from preconception through postpartum, adapting to modern healthcare demands. Midwives today offer a wide range of competencies beyond childbirth, including gynecological care, contraception, sexual health education, and menopausal support, positioning them as essential primary healthcare providers. They collaborate closely with other healthcare professionals to ensure holistic, woman-centered care and use technology for telehealth, monitoring high-risk pregnancies, and providing accessible health education through mobile platforms. Emphasizing mental health, midwives also address perinatal mood disorders and offer psychological support. This field integrates evidence-based practices and fosters continuous education to maintain high care standards and support women's reproductive rights. As midwifery continues to evolve, collaboration with healthcare providers, policymakers, and communities remains crucial to providing empathetic and comprehensive care throughout women's reproductive journeys.

Haider, R., & Das, G. K. (2024). Midwifery of the Future: A Widening Field of Competences. *J. Obstetrics Gynecology and Reproductive Sciences*, 8(1).

- This study assessed the implementation of a digital emergency obstetric and newborn care (EmONC) register in labor wards of selected health facilities in Bangladesh, transitioning from a paper-based system. Developed collaboratively with stakeholders, this Android-based e-register significantly improved usability, data accuracy, and accessibility, achieving a high usability score of 83.7 and acceptance rate over 95%. Adoption was 98% with data fidelity at 90%, indicating that over 80% of relevant data elements were accurately recorded. The system showed utility in 89% of cases and exceeded benchmarks in all outcome metrics. Features like mandatory fields, error-checking, and real-time access streamlined data collection, reduced errors, and enabled timely decision-making. This digital solution addresses common challenges in paper-based systems, such as data delays and loss, by supporting continuous monitoring and reducing the risk of incomplete or missing data. The

successful implementation underscores the potential of digital health records to enhance maternal and newborn care quality, particularly in resource-limited settings like Bangladesh.

Jabeen, S., Rahman, M., Siddique, A. B., Hasan, M., Matin, R., Rahman, Q. S. U., ... & Rahman, A. E. (2024). Introducing a digital emergency obstetric and newborn care register for indoor obstetric patient management: An implementation research in selected public health care facilities of Bangladesh. *Journal of Global Health*, 14.

- Recent studies in Sri Lanka highlight the challenges faced by Public Health Midwives (PHMs) who handle extensive manual documentation, impacting service quality in maternal and child health. A proposed web-based system aims to streamline these processes by digitizing records, simplifying documentation, and improving data management. This system enables midwives to manage daily tasks, track and evaluate field activities, and automatically notify mothers about clinic visits, vaccinations, and prenatal care via multilingual SMS. Additionally, it allows midwives to access and update information on any device, enhancing mobility and efficiency. Midwives have responded positively to the system, finding it practical and supportive for their daily activities, suggesting it as an effective tool for improving healthcare delivery at the community level in Sri Lanka.

Dissanayake, D. W. G. M. L., Vidanagama, D. U., & Wedasinghe, N. (2020). Web-Based Workload Maintain System for Midwives in Sri Lanka.

- This study investigated midwives' competencies and service needs for using mobile health (mHealth) applications to support Midwifery Continuity of Care (MCOC) in West Java. Conducted between August and December 2021, the study combined quantitative and qualitative methods, assessing 373 midwives' characteristics and competencies and engaging 13 midwives in focus group discussions. Most midwives were under 35, had over 10 years of experience, and frequently used smartphones. Findings indicated a strong need for competency development in MCOC, especially in early risk detection and management of complications. Midwives expressed interest in understanding the purposes, benefits, and design of mHealth for MCOC. The study concludes that mHealth applications could enhance MCOC by focusing on early

detection skills, though further research is needed to assess midwives' competencies in applying mHealth within MCOC frameworks.

Susanti, A. I., Ali, M., Hernawan, A. H., Rinawan, F. R., Purnama, W. G., Puspitasari, I. W., & Stellata, A. G. (2022). Midwifery Continuity of Care in Indonesia: Initiation of Mobile Health Development Integrating Midwives' Competency and Service Needs. *International journal of environmental research and public health*, 19(21), 13893.

- The research focused on addressing maternal and child health (MCH) issues in Indonesia, where high mortality rates necessitate better intervention and monitoring. It identified that the recording and reporting of MCH programs at Jagir and Wonokromo Health Centers were inadequate due to manual processes. The study aimed to develop an integrated information system, resulting in the SISKIA (Maternal and Child Health Information and Surveillance System) application, built using the Scrum Management Framework. SISKIA enhances the management of MCH data, particularly for high-risk cases, enables real-time reporting by midwives, and accurately maps health conditions. The application fosters community engagement in reducing maternal and infant mortality rates (MMR and IMR) by providing timely health services, ultimately proving effective in obtaining regular and up-to-date MCH data for improved health outcomes.

Masruroh, N., Fasya, A. H. Z., Handayani, D., & Santoso, A. P. R. (2022). Recording and reporting information system model integrated from midwife practices to public health center. *Bali Medical Journal*, 11(2), 900-904.

- The study addressed the challenges faced by pregnant women during the Covid-19 pandemic, particularly regarding limited mobility and restricted access to health service centers (Puskesmas) for routine visits. Recognizing that pregnant women are at a high risk of virus exposure and that their health conditions could not be adequately monitored, the researchers designed a mobile application called Salute Bidan to serve as a digital guide for monitoring pregnant women's health during the pandemic. Utilizing the ADDIE Framework and the Experiment and Revision Collaboration (ERC) method for software development, Salute Bidan digitizes various midwife tasks related to pregnancy. A key feature of the application is its self-screening tool, developed from 30 risk criteria based on the Poedji Rochyati scorecard, allowing users to assess their pregnancy risk levels. The research produced

both a mobile and a web application, with the mobile app successfully passing all test cases in black-box testing. Overall, Salute Bidan proves to be an effective tool for monitoring pregnant women's health in the context of pandemic-related mobility restrictions.

Nawangarsi, S., Harahap, R. K., Al Rasyid, H., Herlina, N., Ekowati, E., & Asmarany, A. I. (2022). Design of Mobile Digital Healthcare Application For Pregnant Women Based on Android. *MATRIK: Jurnal Manajemen, Teknik Informatika dan Rekayasa Komputer*, 21(2), 439-450.

Research Problem / Problem Statement

- Broad Topic

Digital technology for improvement in health care delivery, especially maternal and newborn care.

- Narrowed Topic

To design a web application with the intention of supporting midwives in managing and simplifying daily tasks related to maternal and newborn care, in such a way that working processes, communication, and data quality will be improved.

- Research Problem

Midwives play a very important role in maternal and newborn healthcare, with a major role in rural and underserved areas. However, they have to overcome enormous challenges in dealing with administrative and clinical tasks that are very important to guarantee the health of mothers and newborns. The tasks involve tracking patient information, appointment management, monitoring of health indicators, and timely intervention. This results in added workload, delayed care, and possible data inaccuracies that impinge on the quality of care. A well-designed digital tool for their needs would alleviate many of the challenges; however, there is a lack of research on how a well-designed digital tool can best be implemented in real clinical practice.

- **Problem Statement**

Administrative burden and challenges in managing the data of maternal and newborn care are continuously growing for midwives, which are affecting the delivery of timely, quality health services. So far, an appropriate, customized web application has not been adopted to ease the daily activities of midwives for higher data accuracy and workflow efficiency. Thus, this study will try to fill this gap by developing a web application that will make the daily activities of midwives easier to manage and reduce their administrative burden so they may work more efficiently toward improving the health outcomes of mothers and newborns.

Objectives of the Study

The main objective of the study will be to design and develop a web application to make midwives' daily tasks easier in handling maternal and newborn care. This study will, therefore, be addressing the major challenges that midwives face in handling patient records, visit scheduling, pregnancy tracking, and communication with other healthcare professionals. In addressing these areas, the research study will establish the particular needs that midwives have in terms of workflow management, and will establish ways through which digital solutions can be adapted in an effective manner to increase their efficiency.

In addition, the research seeks to ensure that the proposed web application is friendly and accessible to midwives irrespective of their level of digital literacy. As such, it will be designed based on the principles of Human-Computer Interaction to ensure friendliness in use and ease of navigation within the system. The research will also establish the validity of the application's functionality in a naturalistic setting with respect to its effect on midwives' multitasking skills and collaboration with other healthcare providers.

The ultimate goal of this study will, therefore, be to ascertain how the web application enhances quality of care to mothers and newborns through timely interventions, proper record-keeping, and coordination among health teams. Through these objectives, the research will contribute to developing a digital tool that reduces midwives' workloads and improves the health outcomes of mothers and newborns.

Significance of the Study

This research study provides a practical solution to the burgeoning administrative and clinical workload facing midwives as regards maternal and newborn health. As it is an easily accessible web application, this will greatly reduce, on a very large scale, the time used by midwives for documentation, scheduling, and health data tracking. In turn, this allows midwives to spend more time with patients in direct care. It will ease midwives' work through the management of patient records, health monitoring, and follow-ups. This, in turn, will have the ultimate effect of increasing the overall health status of mothers and newborn babies—especially in the under-resourced or resource-limited areas of the world where midwives are the most available service providers. It addresses broader challenges in healthcare delivery, such as the digital divide and burgeoning demands for more efficient healthcare solutions. In an increasingly fast-paced world shifting toward leveraging digital tools to enhance healthcare globally, this also calls for increased priority to make sure midwives, especially those in rural areas, are equipped with technology supportive of their capacity for quality healthcare delivery. The study also relates to the global maternal and child health initiatives, such as the SDGs; hence, this study responds to SDG 3: Ensure healthy lives and promote well-being for all at all ages. This web app could help reduce maternal and neonatal mortality by improving the workflow and data management processes of midwives. The timing is everything for this study, as increasingly, global healthcare systems are adopting digital tools to keep up with the rising demands of patients and the increasing need for improved data management. The COVID-19 pandemic accelerated this trend toward telemedicine and digital health solutions, with an evident revival of technology to ensure continuity of care. With strained health services and shortages among personnel, maternal and newborn health services need to be equipped with digital tools that improve the efficacy of midwives; thus, the timing of the adoption is quite apt, considering the present emphasis on strengthening healthcare infrastructures and improving access to quality care in settings with limited resources. It will thus contribute at a timely period to these efforts by providing a practical, scalable digital tool for midwives to make sure they can continue to deliver high-quality maternal and newborn care in a rapidly changing healthcare world.

Methodology of the Study

Study Design

This study will employ a mixed-methods design combining qualitative and quantitative approaches to gather comprehensive insights into midwives' daily tasks, challenges, and

specific requirements for managing maternal and newborn care. The primary objectives are to identify midwives' needs, design an efficient web application, and evaluate its usability and effectiveness.

1. Needs Assessment Phase (Qualitative Analysis):

- Conduct semi-structured interviews with a representative sample of midwives working in maternal and newborn care. This phase aims to understand their workflows, common challenges, and existing gaps in technology support.
- Collect qualitative data on specific tasks and how a web application might help simplify or streamline these tasks.
- Analyze data thematically to identify common patterns and needs that will inform the application design.

2. Prototype Testing (Iterative Design Process):

- Use insights from the needs assessment phase to guide the initial web application design.
- Engage in rapid prototyping, utilizing feedback from a select group of midwives in an iterative design approach. This will involve several rounds of development, testing, and refinement.
- Focus on user-centered design principles to ensure the application is intuitive and effective for midwives' day-to-day tasks.

Study Setting

The study will take place in public health centers, clinics, and hospitals** where midwives provide maternal and newborn care services. This setting is chosen to ensure the application's design aligns with actual work environments and addresses location-specific needs and constraints.

Additionally, the study will consider different geographical regions and levels of resource availability, ensuring the solution's adaptability to varied contexts.

Time Frame/ Time Span

